Unit force:

The amount of force required to produce an acceleration of 1 ms^{-2} in a body of mass 1 kg is called '*unit force*'.

Gravitational unit of force:

In the SI system of units, gravitational unit of force is kilogram force, represented by kg f. In the CGS system its unit is gram force, represented by g f.

1 kgf = 1 kg × 9.8 ms^{-2} = 9.8 N;

1 gf = 1 g × 980 cms^{-2} = 980 dyne

1.6 Impulse

A large force acting for a very short interval of time is called as '**Impulsive force**'. When a force F acts on a body for a period of time t, then the product of force and time is known as '**impulse**' represented by 'J'

$$\text{Impulse, } J = F \times t \qquad (1.7)$$

By Newton's second law

$$F = \Delta p / t \quad (\Delta \text{ refers to change})$$

$$\Delta p = F \times t \qquad (1.8)$$

From 1.7 and 1.8

$$\mathbf{J = \Delta p}$$

Impulse is also equal to the magnitude of change in momentum. Its unit is $kgms^{-1}$ or Ns.

Change in momentum can be achieved in two ways. They are:

i. a large force acting for a short period of time and

ii. a smaller force acting for a longer period of time.

Examples:

- Automobiles are fitted with springs and shock absorbers to reduce jerks while moving on uneven roads.
- In cricket, a fielder pulls back his hands while catching the ball. He experiences a smaller force for a longer interval of time to catch the ball, resulting in a lesser impulse on his hands.



Figure 1.6 Example of impulsive force

1.7 NEWTON'S THIRD LAW OF MOTION

Newton's third law states that **'for every action, there is an equal and opposite reaction. They always act on two different bodies'.**

If a body A applies a force F_A on a body B, then the body B reacts with force F_B on the body A, which is equal to F_A in magnitude, but opposite in direction. $F_B = -F_A$

Examples:

- When birds fly they push the air downwards with their wings (Action) and the air pushes the bird upwards (Reaction).
- When a person swims he pushes the water using the hands backwards (Action), and the water pushes the swimmer in the forward direction (Reaction).
- When you fire a bullet, the gun recoils backward and the bullet is moving forward (Action) and the gun equalises this forward action by moving backward (Reaction).

1.8 PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM

There is no change in the linear momentum of a system of bodies as long as no net external force acts on them.

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